

JSPM University Pune
Faculty of Science and Technology



Lab Manual
“Fundamentals of Python Programming”
Course Code: 240GCSB62_01

Experiment No. 1

- a) Display your name, count length and display what will be your age in year.
- b) Write python program to calculate arithmetic operation using type conversion.

Aim: Basic operations on numbers and strings using python.

Learning Objectives:

- To understand the basics about python programming.
- To learn how to write and execute python program.

Learning Outcomes:

- Upon completing this lab exercise, the learner will be able to
- Write a basic python program.
- Interpreted and run python program.
- Understand the identifiers and operators in python.

Theory:

1. What is Python?
2. Why is python called as dynamic typing language?
3. How to store different types of data inside python identifier?
4. What is the difference between interpreter and compiler?
5. What is IDLE?
6. What is mean by comment in python and how many types of comments?
7. What is mean by type conversion in python?

Algorithm:

- a) Algorithm for display your name, count length and display what will be your age in year.

Step 1: Start

Step 2: Accept the name from the user.

Step 3: Find out the length of that name.

Step 4: Accept your birth year from the user.

Step 5: Find the age = current year - birth_year.

Step 6: Display the name along with its length and age in year.

Step 7: Stop.

b) Algorithm for write python program to calculate arithmetic operation using type conversion.

Step 1: Start

Step 2: Accept the two numbers num1 and num2 from user.

Step 3: Find out the addition of these two numbers ($\text{sum} = \text{num1} + \text{num2}$)

Step 4: Display the addition of two numbers

Step 5: Find out the subtraction of these two numbers ($\text{sub} = \text{num1} - \text{num2}$)

Step 6: Display the subtraction of two numbers

Step 7: Find out the multiplication of these two numbers ($\text{mul} = \text{num1} * \text{num2}$)

Step 8: Display the multiplication of two numbers

Step 9: Find out the division of these two numbers ($\text{div} = \text{num1} / \text{num2}$)

Step 10: Display the division of two numbers

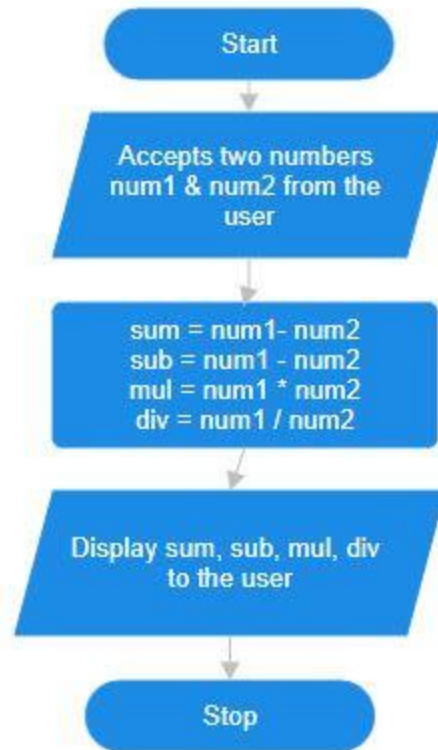
Step 11: Stop.

Flowchart:

a) Flowchart for display your name, count length and display what will be your age in year.



b) Flowchart for python program to calculate arithmetic operation using type conversion.



Conclusion: Hence, we learn how to execute the python program through command line argument and also learn how to accept input from the user and how to process this input with basic arithmetic operations.

Experiment No. 2

Problem Statement: Write simple python program that demonstrates the use of lists, tuples, and dictionaries

Aim: To learn basic data structures of python.

Learning Objectives:

- To understand the basics about python data structures.
- To learn how to create and manipulate data inside lists, tuples and dictionaries.

Learning Outcomes:

- Upon completing this lab exercise, the learner will be able to
- Write a basic python data structures syntax and semantics.
- How to create empty list and how to perform various operations on lists for data manipulations.
- How to create empty tuple and how to perform various operations on tuple for data manipulations.
- How to create empty dictionary and how to perform various operations on dictionary for data manipulations.

Theory:

1. What are the mutable and immutable objects in python?
2. How to create and access elements from lists?
3. How to modify and slice the lists elements?
4. What is mean by nested list explain with example?
5. What type of data we can store in list?
6. Why do we require dictionaries in python?

Algorithm:

Step 1: Start

Step 2: Create an empty list

Step 3: Create list of string, integer, float number.

Step 4: Create list of different data types and duplicate numbers.

Step 5: Perform Crud operation

Step 6: Stop.

Flowchart:



Conclusion: Hence, we learn how to implement the python basic data structures like list, tuple, dictionary and also learn various operation of those data structures.

Experiment No. 3

Problem Statement: Write a function to calculate factorial of a number. Develop a program to compute binomial coefficient (Given N and R).

Aim: To implement function using python programming.

Learning Objectives:

- To understand the basics about python functions.
- To learn how to create and manipulate data inside functions.
- To learn types of functions.

Learning Outcomes:

- Upon completing this lab exercise, the learner will be able to
- Write a basic python function syntax and semantics.

- To implement functions for various task.
- Clear the difference between user defined function and built in function.

Theory:

1. Why is indenting required in python?
2. Explain with example all decision-making statement in python?
3. Why do we require iteration statement in python?
4. Explain with example while loop in python?
5. Explain with example for loop with break and continue statement in python?
6. What is function? also explain why do we require functions in python programming language?

Algorithm:

Step 1: Start

Step 2: Accept N and R from the user.

Step 3: using function of factorial compute the factorial of N.

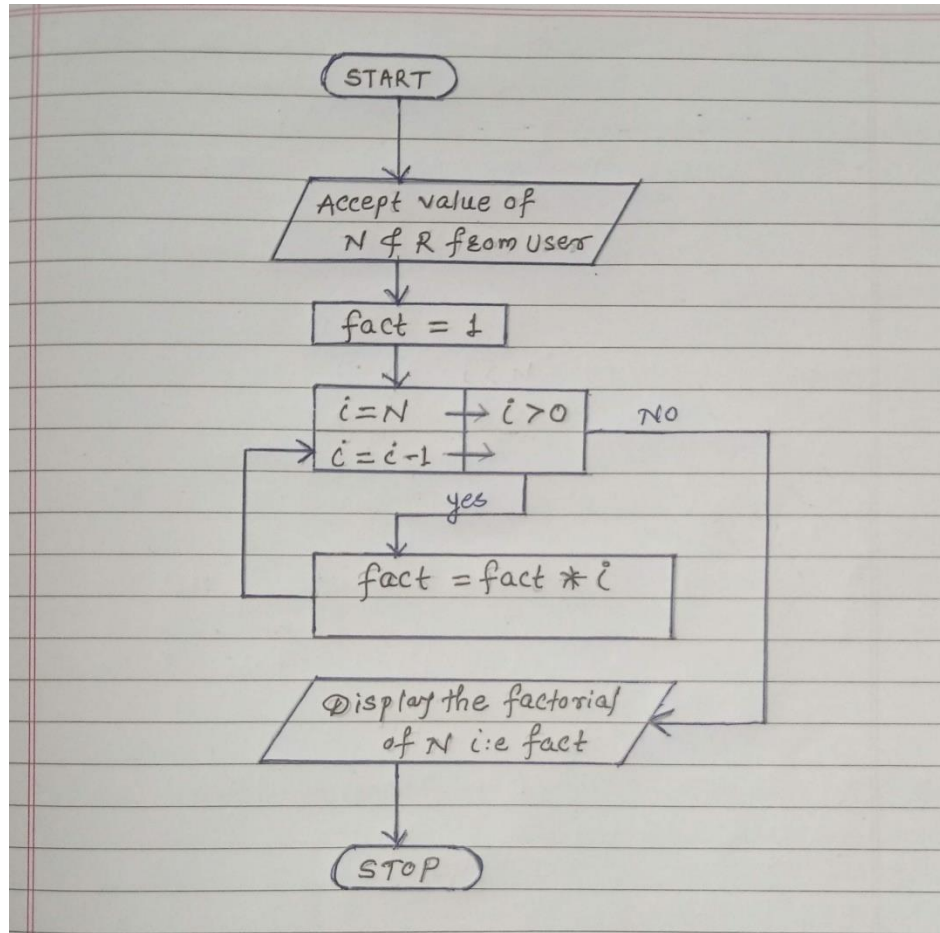
Step 4: Display the factorial of value N.

Step 5: Compute the binomial coefficient of N and R $(N!) / ((N!) * (N-R)!)$

Step 6: Display the binomial coefficient of N and R.

Step 7: Stop.

Flowchart:



Conclusion: Hence, we learn how to implement the function like user defined function for computation of factorial of any number n .

Experiment No. 4

Problem Statement: Develop a Python program to check whether a given number is palindrome or not and also count the number of occurrences of each digit in the input number.

Aim: To understand the uses of looping and decision-making statement.

Learning Objectives:

- To understand the basics about python decision making statement.
- To understand the basics about python looping statements.

Learning Outcomes:

- Upon completing this lab exercise, the learner will be able to
- Write a basic python looping statement syntax and semantics.
- How and when to use decision making statements.

Theory:

1. What is object-oriented programming?
2. What is class and objects explain with example?
3. What encapsulation and abstraction explain with example?
4. What is mean by Inheritance explain with example?

Algorithm:

Step 1: Start

Step 2: Accept number n from user.

Step 3: using user defined function of reverse, reverse the number n and store it into rev identifier.

Step 4: Check number n and rev are equal or not. if equal show palindrome else shoe not palindrome message to the user.

Step 5: Compute the occurance of each digit inside the number n using user defined function.

Step 6: Display the count of occurance of each digit.

Step 7: Stop

Flowchart: Draw this flowchart by your own.

Conclusion: Hence, we learn how to implement the function, looping and decision-making statements to develop a logic to check whether given number is palindrome or not.

Experiment No. 5

Problem Statement: Develop a program to read the student details like name, USN and marks in 3 subjects. Display the student details, total marks and percentage with suitable messages

Aim: To implement class by using student entity.

Learning Objectives:

- To understand difference between procedure-oriented language and object-oriented language.
- To understand the uses of oops features like encapsulation and data hiding.
- To understand the syntax of writing class.

Learning Outcomes:

- Upon completing this lab exercise, the learner will be able to
- Able to map any entity into the class.
- Able to analyze the problem statement.
- Able to identify properties and behavior of entity.

Theory:

1. Explain different types of Inheritance?
2. What is exception handling?
3. What is file? how to perform reading and writing operation on file?
4. Explain seek (), tell (), readline(), readlines() methods.

Algorithm:

Step 1: Start

Step 2: Accept student name, USN and marks of three subjects (FPP, AC, SPM)

Step 3: using function find out the total of three subjects.

Step 4: Using function find out the percentage of students.

Step 5: Display all the details of student.

Step 6: Stop

Flowchart: Draw this flowchart by your own.

Conclusion: Hence, we learn how to implement the class in python along with its object creation and calling of functions as per the priority user requirement.

Experiment No. 6

Problem Statement: Write a Python program that demonstrates the use of a recursive function, nested functions, and functional programming techniques such as map, filter, lambda functions and print the result, which is the sum of the squares of the even number.

Learning Objectives:

- To understand the basics about python inbuilt functions like map, filter and lambda.
- To learn how to write and execute recursive and nested function.

Learning Outcomes:

- Upon completing this lab exercise, the learner will be able to
- Able to identify how and when to use map function.
- Able to identify how and when to use filter function.
- Understand the use of lambda function and its implementation.
- Implementation of recursive function and nested function.

Theory:

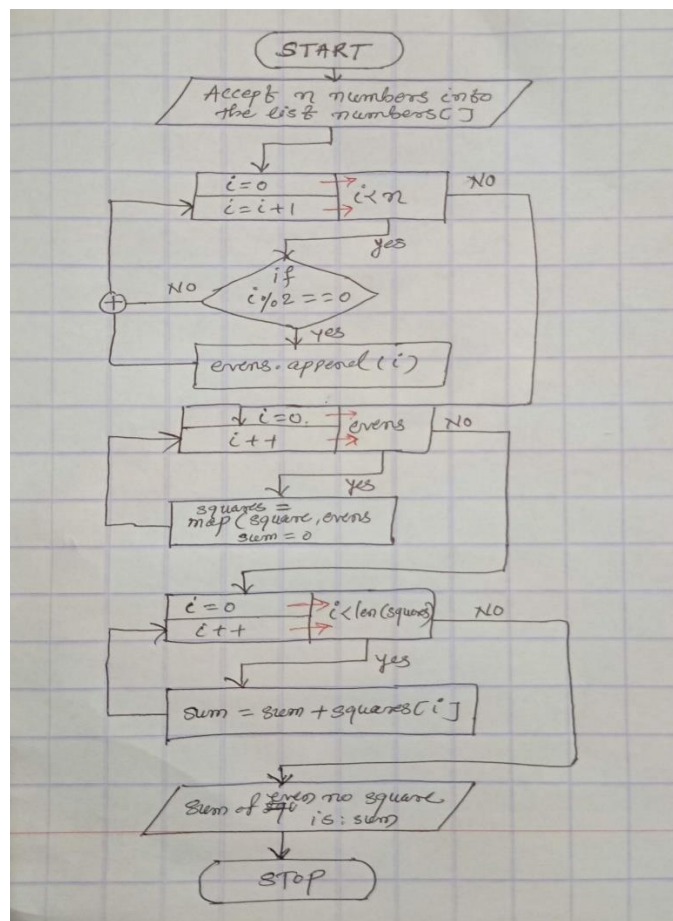
1. Why do we require recursive function explain with example?
2. What is mean by nested function explain with example?
3. What is map () function and why do we require map () function explain with example?
4. What is filter () function and why do we require filter () function explain with example?
5. What is lambda () function and why do we require lambda () function explain with example?

Algorithm:

1. **Step 1:** Start
2. **Step 2:** Accept n numbers from the user and stored it inside numbers [] list.

3. **Step 3:** Pass numbers [] list as an argument to `sum_of_square_of_evens(numbers)`
4. **Step 3.1** First find out the even numbers from the list with the help of filter function.
5. **Step 3.2** Secondly with the help of map function find out the square of all even numbers and stored it into squares.
6. **Step 3.3** Pass this square in list type to recursive_sum function to find out sum of square of even numbers.
7. **Step 4:** Show the sum to user
8. **Step 5:** Stop

Flowchart:



Conclusion: Hence, we learn how to execute the recursive and nested function in python. Also learn how to implement built in function map, filter and lambda in python.

Experiment No. 7

Problem Statement:

Write a python program by creating a class called Employee to store the details of Name, Employee_ID, Department and Salary, and implement a method to update salary of employees belonging to a given department.

Aim: Basic operations on class and its operations.

Learning Objectives:

- To understand the basics about python Classes concept
- To learn how to write and execute python object-oriented program.

Learning Outcomes:

- Upon completing this lab exercise, the learner will be able to
- Write a basic python program using class.
- Able to implement various features of object-oriented programming.
- Understand the memory management of objects.

Theory:

1. Why do we require Inheritance?
2. Explain how to create file in writing mode? also how to write content inside the file and how to read those and print?
3. When to use seek () and tell () method explain with example.?
4. Explain the difference between readline() and readlines() method?

Algorithm:

Step 1: Start

Step 2: Ask user how many **(n)** employees' information he /she wants to store

Step 3: Accept the **n** employee details like employee id, name, department and salary from the user.

Step 4: Display the **n** employee information.

Step 5: Now ask user which department salary he/she wants to update.

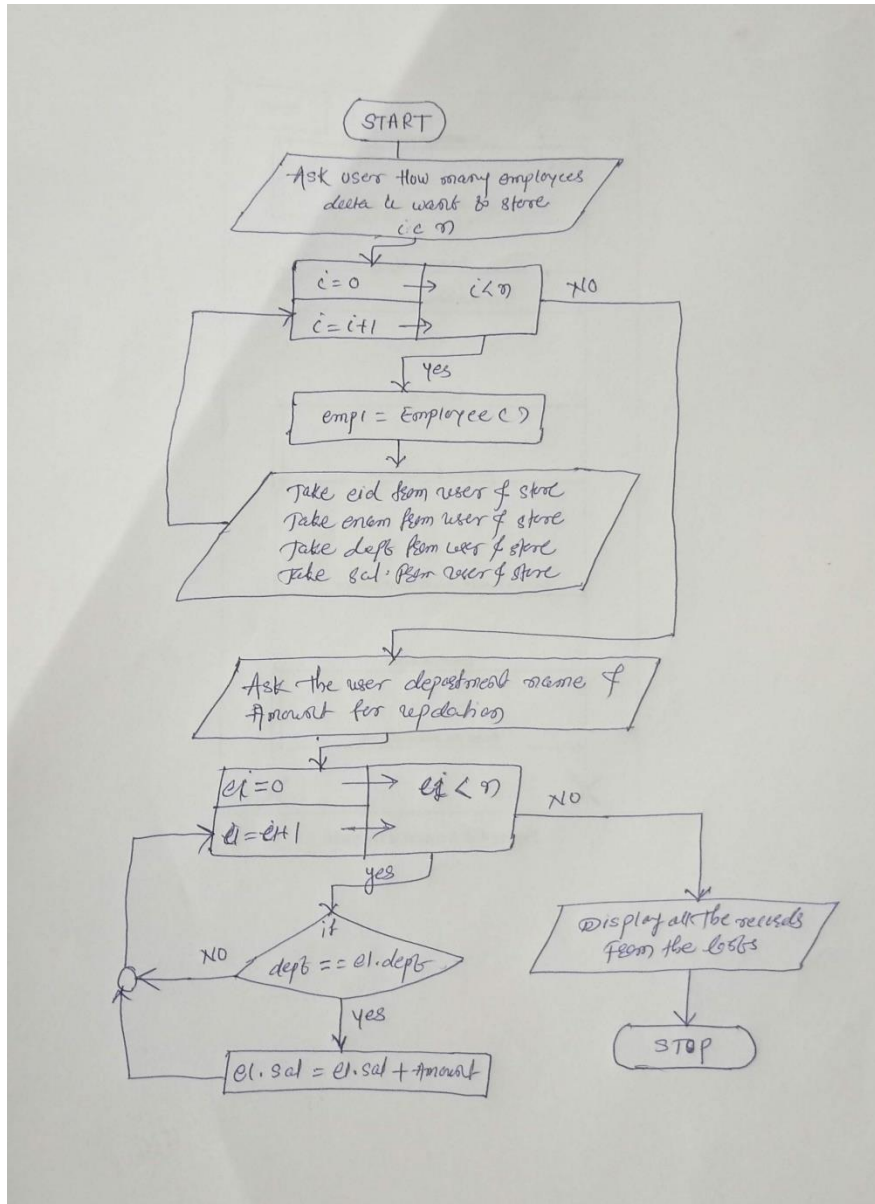
Step 6: Ask user for the amount to update in existing salary.

Step 7: Search the department among **n** available objects and update accordingly.

Step 8: Finally, once again show the updated record of employees.

Step 9: Stop.

Flowchart:



Conclusion: Hence, we learn how to execute the python class and object program through array of objects and also learn how to accept input from the user and how to process this input with basic arithmetic operations.

Experiment No. 8

Problem Statement: Write a python program to accept a file name from the user and perform the following operations

1. Display the first N line of the file.
2. Find the frequency of occurrence of the word accepted from the user in the file.

Learning Objectives:

- To understand the basics about python file handling.
- To learn how to create a file in various mode and how to perform various operation on it.

Learning Outcomes:

- Upon completing this lab exercise, the learner will be able to
- Able to create file, write content in it and close the file.
- Able to open file in read mode, move the file pointer in any location inside the file.
- Understand the difference between data streaming and data buffering.
- Implementation of file program for user specific requirements.

Theory:

1. What is data streaming?
2. What is data buffering?
3. What is Input Output Stream and what is buffer?
4. Explain in detail different file access modes.
5. How to write inside file and how to read one character at a time from file.

Algorithm:

Step 1: Start

Step 2: Accept the file name from the user.

Step 3: We are assuming that file is already created on HDD, so we simply open it into read mode.

Step 4: We are asking user how many n lines you want to display from file.

Step 5: Displaying the first n lines from the file.

Step 6: We are asking user to enter which word occurrence he/she wants to find from the file.

Step 7: Counting the occurrence of that word from the file.

Step 8: Showing the occurrence of that word to the user.

Step 9: Stop

Conclusion: Hence, we learn how to create the file on secondary storage using python programming. also, we learn various operations on file to process the data as per user requirement.

Experiment No. 9

Problem Statement:

Write Python program covers basic functionalities provided by the math module.

Learning Objectives:

- To understand the basics about python modules in python.
- To learn various built-in modules inside python.
- To learn regular expression and its methods in python.

Learning Outcomes:

- Upon completing this lab exercise, the learner will be able to
- Able to create modules inside python.
- Able to create our own library by wrapping various function related with each other.
- Understand the various built-in modules inside python like math module
- Implementation of basic functionalities provided by the math module.

Theory:

1. What is Module in python and how to create it?
2. What is loading and reloading of module?
3. How the module search path work in the python?
4. List any nine methods of math modules.

Algorithm:

- Step 1: Start
- Step 2: Display two constants of math module
- Step 3: Call to basic functions in math modules.
- Step 4: Call three Trigonometric functions from math modules.
- Step 5: Call two Logarithmic functions from math module.
- Step 6: Call two Exponential functions from math module.
- Step 7: Call two Rounding functions from math module.
- Step 8: Call factorial and GCD functions from math module.
- Step 9: Call Degree and Radians functions from math module.
- Step 10: Stop

Conclusion: Hence, we learn all the functionalities inside math module.

